



September 2025

(For IS)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

12. In event-based architecture, the component that announces an event is called:
 A) Listener B) Publisher
 C) Subscriber D) Controller
13. Which style fits sensor-driven robotics?
 A) Process control B) Pipes and filters
 C) Repository D) Interpreter
14. The broker acts as:
 A) Mediator between clients and servers B) Storage system
 C) Event generator D) Compiler
15. A common example of broker pattern:
 A) CORBA B) Layered style
 C) MVC D) PAC
16. Which of these belongs to "Adaptable Systems" category?
 A) Microkernel pattern B) MVC
 C) PAC D) Repository style
17. A key activity in designing software architecture is:
 A) Identifying architectural drivers B) Debugging code
 C) Writing manuals D) Marketing the product
18. Which of the following is NOT an input to designing the architecture?
 A) Quality attribute requirements B) Functional requirements
 C) Business constraints D) Source code formatting rules
19. Which is NOT a type of architectural view?
 A) Module view B) Component-and-connector view
 C) Allocation view D) Color scheme view
20. A deployment view maps:
 A) Software to hardware nodes B) GUI to database
 C) Test cases to code D) Views to users

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the Architecture Business Cycle (ABC). Discuss how business goals, technical environment, and stakeholder needs influence software architecture.	8	L*2	1	1,2
	b) What makes a "good" software architecture? Explain the essential characteristics and qualities of a good architecture with suitable examples.	8	L2	1	1,2
2.	a) Discuss the concept of Architectural Patterns, Reference Models, and Reference Architectures. How do they assist in architectural design?	8	L2	1	1,2
	b) Explain System Quality Attributes. What are quality attribute scenarios and how are they used in practice?	8	L3	2	1,3
3.	a) What are Tactics in Software Architecture? Explain the role of tactics in achieving system quality attributes.	8	L2	2	1,2
	b) Discuss Testability and Usability Tactics. How do they enhance the maintainability and user experience of software systems?	8	L2	2	1,2
Unit – II					
4.	a) Define Architectural Styles. Explain their significance in software architecture with suitable examples.	8	L2	3	1,2
	b) Describe the Process Control architectural style. In what type of systems is it commonly used?	8	L3	3	1,2

IS2314-1**SEE - September 2025**

5.	a)	Discuss Heterogeneous Architectures. How do they integrate multiple styles within a single system? Give an example	8	L2	3	1,2
	b)	Describe the Model-View-Controller (MVC) pattern. Explain how it separates concerns and improves maintainability in interactive systems.	8	L2	4	1,2
6.	a)	Explain the Microkernel architectural pattern. Discuss its components and how it supports adaptability and portability	8	L2	4	1,3
	b)	Discuss the importance of architectural patterns in software design. How do they help in solving recurring architectural problems?	8	L2	4	1,2
Unit – III						
7.	a)	Discuss the importance of forming the team structure. How does team organization impact the success of software architecture?	8	L2	5	1,2
	b)	List and explain the various uses of architectural documentation. Why documentation is considered critical in large-scale software development?	8	L2	5	1,2
8.	a)	Describe how to document a view. What elements should be included to make the documentation complete and useful?	8	L2	5	1,2
	b)	Discuss the challenges and methods of maintaining documentation across multiple views. How can consistency be ensured.	8	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
